



Taylor's Theorem and Mean Value Theorem for Random Functions and Random Variables

Yifan Yang, Xiaoyu Zhou & Ming Wang

To cite this article: Yifan Yang, Xiaoyu Zhou & Ming Wang (21 May 2026): Taylor's Theorem and Mean Value Theorem for Random Functions and Random Variables, The American Statistician, DOI: [10.1080/00031305.2026.2657484](https://doi.org/10.1080/00031305.2026.2657484)

To link to this article: <https://doi.org/10.1080/00031305.2026.2657484>



View supplementary material [↗](#)



Published online: 21 May 2026.



Submit your article to this journal [↗](#)



Article views: 136



View related articles [↗](#)



View Crossmark data [↗](#)

Taylor's Theorem and Mean Value Theorem for Random Functions and Random Variables

Yifan Yang^a , Xiaoyu Zhou^b, and Ming Wang^a 

^aDepartment of Population and Quantitative Health Sciences, Case Western Reserve University, Cleveland, OH; ^bLaboratory of Cognition & Emotion, University of Maryland, College Park, MD

ABSTRACT

This study addresses the often-overlooked issue of measurability at intermediate points when applying Taylor's theorems to random functions and random vectors (e.g., likelihood functions with respect to estimators) in statistics. Classical Taylor-related theorems were originally developed for deterministic settings. Consequently, they do not directly extend to stochastic functions and variables and do not inherently guarantee the measurability of intermediate points. In statistical contexts, applying these theorems without properly accounting for randomness can lead to analyses that lack well-defined probabilistic interpretations. Elementary approaches, such as pointwise constructions, are insufficient for handling random quantities and establishing measurable intermediate points. Moreover, some statistical literature has implicitly disregarded this issue, often neglecting the stochastic nature of the problem and assuming that intermediate points are measurable. To address this gap, we develop multivariate Taylor's and mean value theorems tailored for random functions and random variables under mild assumptions. We provide illustrative examples demonstrating the applicability of our results to commonly used statistical methods, including maximum likelihood estimation, M -estimation, and profile estimation. Our findings contribute a rigorous foundation for the applications of Taylor expansions in statistics.

ARTICLE HISTORY

Received April 2025
Accepted April 2026

KEYWORDS

Filippov implicit function theorem; Measurable Taylor's theorem; quasi-Carathéodory function; quasi-continuous function

1. Introduction

Taylor-related theorems, including Taylor's theorem, Taylor series, and the mean value theorem (i.e., Taylor's theorem for the first-order derivative), are fundamental tools in calculus for analyzing the local properties of functions. Taylor polynomials provide linear approximations of complex functions, as originally formulated by Taylor (1717). Classical versions of these theorems, along with their extensions to Banach spaces, are detailed in Lang (1993). For clarity and ease of comparison, we present a classical version of the mean value theorem below:

(Deterministic mean value theorem) Suppose $\mathbf{x}$ and $\mathbf{y}$ are deterministic vectors, and f is a deterministic function satisfying certain smoothness conditions. Then, there exists a deterministic vector ξ such that $f(\mathbf{x} + \mathbf{y}) = f(\mathbf{x}) + \nabla f(\xi) \mathbf{y}$, where ∇f is the row vector of the gradient of f .

It is important to emphasize that the above classical mean value theorem (and the classical Taylor's theorem) follows a “deterministic in, deterministic out” principle: it requires $\mathbf{x}$, $\mathbf{y}$, and f to be deterministic and yields a deterministic intermediate point ξ .

In statistics, the mean value theorem and Taylor's theorem are widely used to establish important results, such as the Delta method and the asymptotic properties of likelihood-based estimators. However, in this context, both functions and variables are typically constructed from the random sample, making them inherently random rather than deterministic.

Consequently, directly applying classical Taylor's theorems to these random functions and variables can be problematic and may lack mathematical rigor due to neglecting considerations of randomness.

To illustrate this issue, consider an example in establishing the asymptotic normality of the sample mean estimator $\hat{\theta}$ based on a single observation $X \sim \text{Poisson}(\theta)$, where the population parameter $\theta > 0$ is unknown. The probability mass function of X is $p(x; \theta) = \theta^x \exp(\theta) / x!$, resulting in the log-likelihood function (for a single observation): $\ell(\theta; x) = -\theta + x \log(\theta) - \log(x!)$. The score function with respect to θ is given by $\ell^{(1)}(\theta; x) = -1 + x/\theta$ and its derivative is $\ell^{(2)}(\theta; x) = -x/\theta^2$. To explore the asymptotic properties of $\hat{\theta}$, we replace the realization x with the random variable X in the log-likelihood function and consider the second-order Taylor expansion of the random log-likelihood function $\ell(\cdot; X)$, evaluated at $\hat{\theta}$, around the true value θ :

$$\begin{aligned} \ell(\hat{\theta}; X) &= \ell(\theta; X) + (\hat{\theta} - \theta) \ell^{(1)}(\theta; X) \\ &\quad + \frac{1}{2} (\hat{\theta} - \theta)^2 \ell^{(2)}(\tilde{\theta}; X), \end{aligned}$$

where $\tilde{\theta}$ is a “point” between $\hat{\theta}$ and θ . Two key questions arise regarding the above application of the classical Taylor's theorem:

1. Does the classical Taylor's theorem remain valid for the random function $\ell(\hat{\theta}; X) = -\hat{\theta} + X \log(\hat{\theta}) - \log(X!)$ and the random variable $\hat{\theta} \triangleq \hat{\theta}(X)$?

2. Can the classical Taylor's theorem justify that the intermediate point $\tilde{\theta}$ is measurable?

Unfortunately, both answers are “no” due to the “deterministic in, deterministic out” principle of the classical theorems. These issues are crucial in statistical applications. The randomness of the likelihood function $\ell(\hat{\theta}; X)$ and the estimator $\hat{\theta}(X)$ poses challenges in rigorously applying Taylor-related theorems in statistical contexts, potentially undermining subsequent probabilistic arguments concerning statistical properties, such as consistency (i.e., estimators converge in probability) or asymptotic normality (i.e., estimators converge in distribution).

To address this limitation, we propose a random version of the mean value theorem (and Taylor's theorem) that follows a “random in, random out” principle:

(Random mean value theorem) Suppose X , Y , and Z are random vectors and $f(Z, \cdot)$ is a random function satisfies certain smoothness conditions. Then, there exists a random vector ξ such that $f(Z, X + Y) = f(Z, X) + \nabla f(Z, \xi) Y$, where ∇f is the row vector of the gradient of f (with respect to its second argument).

However, proving this measurable version of Taylor-related theorems is nontrivial. Elementary approaches, such as a “pointwise” argument and a “function composition” argument, fail to establish measurability, as demonstrated by counterexamples constructed using a non-measurable set. For ease of presentation, we set $p = 1$, assume function $f(Z; \cdot) = f(\cdot)$ is deterministic and only X and Y are random variables. In this setting, consider the “pointwise” argument: for every $\omega \in \Omega$, $X(\omega)$ is a deterministic number. The classical mean value theorem guarantees the existence of a deterministic number ξ_ω (which may not be unique), located on the segment between $X(\omega)$ and $X(\omega) + Y(\omega)$, such that $f(X(\omega) + Y(\omega)) = f(X(\omega)) + \nabla f(\xi_\omega) Y(\omega)$. Even if we assume uniqueness for each ω and allow ω to vary over the entire Ω , the resulting pointwise-defined map $\xi : \Omega \rightarrow \mathbb{R}, \omega \mapsto \xi_\omega$ does not necessarily ensure measurability. A counterexample involves the indicator function on a non-measurable set: let $\Omega = \mathbb{R}$ and $N \subseteq \Omega$ be a non-measurable set. Then the pointwise-defined indicator function $\mathbb{I}_N : \Omega \rightarrow \{0, 1\}$, which takes the value 1 if $\omega \in N$ and 0 otherwise, is not measurable. Another approach involves an implicit “function composition” argument: $\nabla f(\xi) = \{f(X + Y) - f(X)\} Y^{-1}$. One might incorrectly argue that ξ is measurable based on the following false claim: if both the function g (e.g., ∇f) and the composition $g(h)$ (e.g., $\{f(X + Y) - f(X)\} Y^{-1}$) are measurable, then the function h (e.g., ξ) is measurable. Unfortunately, this claim is false. A counterexample is the function $h : [-1, 1] \rightarrow \{-1, 1\}$, which takes the value 1 on a non-measurable subset of $[-1, 1]$ and -1 otherwise. Let $g : [-1, 1] \rightarrow [0, 1], x \mapsto |x|$ be the absolute value function, which is continuous and therefore measurable. We find that $g(h)(x) \equiv 1$ is measurable, while g is measurable, but h is not.

Existing literature provides valuable insights, but does not directly resolve the randomness and the measurability issues. For example, Benichou and Gail (1989) applied the implicit function theorem to define random vectors implicitly, and Massey and Whitt (1993), Lin (1994), and Di Crescenzo (1999) developed probabilistic versions of Taylor expansions under expectation. However, these approaches impose restrictions, such as requiring nonnegative random variables with finite

means. Aliprantis and Border (2006) established a univariate version of Taylor's theorem for a deterministic function and a random variable, demonstrating that a measurable assignment $\xi(\omega)$ exists. Despite these contributions, the literature remains largely fragmented.

In this article, we propose the multivariate Taylor-related theorems for random functions and random variables and provide simplified proofs under the mild assumptions. The proposed results address the randomness and the measurability gaps in statistical applications, offering a rigorous foundation for many Taylor-based proofs in statistics. The rest of the article is organized as follows. Section 2 introduces the proposed Taylor's theorems in two scenarios: deterministic functions and random variables, and random functions and random variables. Section 3 discusses statistical applications, presenting a statistical version of Taylor-related theorems for random log-likelihood functions and estimators. Additional discussions are presented in Section 4.

2. Main Results

We start with the classical notation provided in Lang (1993). Suppose $\mathcal{E}$ and $\mathcal{F}$ are two Banach spaces. Let $\mathcal{U} \subseteq \mathcal{E}$ be an open set, and let $f : \mathcal{U} \rightarrow \mathcal{F}$ be of class C^m for a given integer $m \geq 1$, that is, f and $f^{(k)}$ are continuous on $\mathcal{U}$ for $k = 0, 1, \dots, m$. Here, $f^{(k)}$ denotes the k th order derivative of f for integer $k \geq 0$ (for convenience, we define $f^{(0)} \triangleq f$ and $0! \triangleq 1$). Furthermore, let $x \in \mathcal{U}$ and $y \in \mathcal{E}$ such that $x + ty \in \mathcal{U}$ for all $t \in [0, 1]$. Then, we have

$$f(x + y) = \sum_{k=0}^{m-1} \frac{1}{k!} f^{(k)}(x) y^{(k)} + R_m, \\ R_m \triangleq \int_0^1 \frac{(1-t)^{m-1}}{(m-1)!} f^{(m)}(x + ty) y^{(m)} dt, \quad (1)$$

where $y^{(k)}$ represents the k -tuple $(y, \dots, y)$ for integer $k \geq 1$ (Lang 1993), and R_m is the remainder in integral form. The deterministic versions of Taylor's and the mean value theorems are formally provided in the [Supplementary Material](#). Throughout the article, let $\Sigma_{\mathcal{A}}$ and $\mathcal{B}_{\mathcal{A}}$ denote the σ -algebra and Borel σ -algebra of subsets of set $\mathcal{A}$, respectively.

2.1. Deterministic Functions

In this section, we introduce [Theorem 1](#), our first extension of the classical Taylor's theorem. Unlike the classical setting where inputs are deterministic, we replace the deterministic vectors x and y with random vectors X and Y , while the function f remains deterministic.

Theorem 1 (Taylor's theorems for Deterministic Functions and Random Vectors). Let (Ω, Σ_Ω) , $(\mathcal{X}, \mathcal{B}_{\mathcal{X}})$, $(\mathcal{Y}, \mathcal{B}_{\mathcal{Y}})$, and $(\mathcal{U}, \mathcal{B}_{\mathcal{U}})$ be measurable spaces, where $\mathcal{U} \subseteq \mathcal{E} = \mathbb{R}^p$, $\mathcal{F} = \mathbb{R}$, $\mathcal{X} \subseteq \mathcal{U}$, $\mathcal{Y} \subseteq \mathcal{E}$, and $m \geq 1$ is an integer. Let $f : \mathcal{U} \rightarrow \mathcal{F}$ be a deterministic function. Suppose the following three conditions hold:

(D-1) $\mathcal{U}$ is an open set.

(D-2) $f(\cdot)$ is m th continuously differentiable on $\mathcal{U}$.

(D-3) Let $X : (\Omega, \Sigma_\Omega) \rightarrow (\mathcal{X}, \mathcal{B}_\mathcal{X})$ and $Y : (\Omega, \Sigma_\Omega) \rightarrow (\mathcal{Y}, \mathcal{B}_\mathcal{Y})$ be p -dimensional random vectors such that $\{X(\omega) + tY(\omega) : t \in [0, 1]\} \subseteq \mathcal{U}$ for all $\omega \in \Omega$.

Then, there exists a p -dimensional random vector $\xi : (\Omega, \Sigma_\Omega) \rightarrow (\mathcal{U}, \mathcal{B}_\mathcal{U})$ such that

$$f(X + Y) = \sum_{k=0}^{m-1} \frac{1}{k!} \left(\sum_{\ell=1}^p Y_\ell \frac{\partial}{\partial u_\ell} \right)^k f(X) + \frac{1}{m!} \left(\sum_{\ell=1}^p Y_\ell \frac{\partial}{\partial u_\ell} \right)^m f(\xi),$$

and $\xi(\omega) \in \{X(\omega) + tY(\omega) : t \in [0, 1]\} \subseteq \mathcal{U}$ for each $\omega \in \Omega$, where u_ℓ and Y_ℓ are the ℓ th components of $\mathbf{u}$ and $\mathbf{Y}$, and the operator is short for

$$\left(\sum_{\ell=1}^p Y_\ell \frac{\partial}{\partial u_\ell} \right)^k = \sum_{i_1, \dots, i_p \in \mathbb{N} : \sum_{\ell=1}^p i_\ell = k} \frac{k!}{i_1! \dots i_p!} Y_1^{i_1} \dots Y_p^{i_p} \frac{\partial^k}{\partial u_1^{i_1} \dots \partial u_p^{i_p}}, \quad k = 0, 1, \dots, m-1.$$

In particular, for $m = 1$, the mean value theorem holds: there exists a p -dimensional random vector $\xi : (\Omega, \Sigma_\Omega) \rightarrow (\mathcal{U}, \mathcal{B}_\mathcal{U})$ such that

$$f(X + Y) - f(X) = \nabla f(\xi) Y,$$

and $\xi(\omega) \in \{X(\omega) + tY(\omega) : t \in [0, 1]\} \subseteq \mathcal{U}$ for each $\omega \in \Omega$, where $\nabla f(\mathbf{u}) \triangleq (\partial f(\mathbf{u})/\partial u_1, \dots, \partial f(\mathbf{u})/\partial u_p)$ is the gradient (row) vector.

Although this generalization simply replaces deterministic vectors with random ones, the mathematical proof is complex and is detailed in the [Supplementary Material](#). The proof relies on the classical Taylor's theorem but incorporates the Filippov implicit function theorem (Himmelberg 1975) to handle the remainder term. Essentially, the Filippov theorem acts as a “measurable selection theorem”, ensuring that a suitable measurable function exists. We further show in [Corollary 1](#) that this result extends naturally to a class of quasi-continuous functions.

Theorem 1 extends the Taylor expansion to deterministic functions evaluated at random vectors. Compared to the classical deterministic theorem, there are two key distinctions. First, regarding assumptions, condition (D-3) specifically requires the input vectors to be random. Second, regarding the results, [Theorem 1](#) explicitly establishes that the intermediate point is random. This is a crucial addition, as the classical theorem cannot guarantee the randomness of this point. A direct application of [Theorem 1](#), the proof of the Delta method, is presented in [Section 3.1](#).

Remark 1 (Univariate, Multivariate, and Vector-Valued Cases). [Theorem 1](#) presents the multivariate version of extended Taylor-related theorems for $p \geq 1$. When $p = 1$, it naturally reduces to the univariate case. Additionally, let $\mathcal{E} = \mathbb{R}^p$ and $\mathcal{F} = \mathbb{R}^q$, where $p \geq 1$ and $q \geq 1$ are integers. In this vector-valued setting, the classical Taylor's Theorem can still be formulated in the integral form (1), where $f(\mathbf{u}) \triangleq (f_1(\mathbf{u}), \dots, f_q(\mathbf{u}))^\top$:

$\mathbb{R}^p \rightarrow \mathbb{R}^q$ and each component $f_r(\mathbf{u}) \triangleq f_r(u_1, \dots, u_p) : \mathbb{R}^p \rightarrow \mathbb{R}$ for $r = 1, \dots, q$. Higher-order derivatives are computed iteratively via the first-order derivative matrix, given by $\nabla f(\mathbf{u}) = (\partial f_r(\mathbf{u})/\partial u_\ell)_{r=1, \ell=1}^{q,p} \in \mathbb{R}^{q \times p}$. However, unlike the scalar case, there is no direct vector-valued analogue of the deterministic mean value theorem (and consequently, no such result in a measurable sense). It is because there does not exist a universal ξ lying between $\mathbf{x}$ and $\mathbf{y}$ (applicable simultaneously across all components of f) such that $f(\mathbf{x} + \mathbf{y}) - f(\mathbf{x}) = \nabla f(\xi) \mathbf{y}$, as discussed in Tao (2011, 2022). This limitation has led to occasional misapplications in the statistical literature. A solution, which uses Taylor's theorem in its integral form, is outlined in Feng et al. (2013).

Remark 2 (Continuity of $f^{(m)}$). Certain versions of the classical Taylor's theorem require only the existence of $f^{(m)}$ on $\mathcal{U}$ without assuming its continuity (Rudin 1976). In contrast, [Theorem 1](#) imposes the continuity condition (D-2) on $f^{(m)}$. Later in the article, we relax this continuity assumption.

Building on [Remark 2](#), we present the following extension of [Theorem 1](#), where $f^{(m)}$ is not continuous on $\mathcal{U}$.

Corollary 1 (Taylor's Theorems for Quasi-Continuous Functions with Closed Graphs). Suppose conditions (D-1) and (D-3) in [Theorem 1](#), along with the following condition hold:

(D-2') $f(\cdot)$ is $(m-1)$ th continuously differentiable and $f^{(m)}(\cdot)$ is quasi-continuous with a closed graph on $\mathcal{U}$.

Then, the conclusions in [Theorem 1](#) still hold.

The definition of quasi-continuous functions and a detailed proof are provided in the [Supplementary Material](#). [Corollary 1](#) ensures the measurability of intermediate points for quasi-continuous functions with closed graphs, even when they are discontinuous. An example is the piecewise continuous function $f(x) = \begin{cases} 1/x, & x > 0 \\ 0, & x \leq 0 \end{cases}$, which is discontinuous but is quasi-continuous with a closed graph (Dindoš and Toma 1997).

2.2. Random Functions

Our primary goal is to develop a “statistical version” of Taylor's theorem (i.e., [Theorem 3](#)) that applies directly to statistical problems, such as exploring the asymptotic behavior of maximum likelihood estimators. To achieve this, we replace the deterministic function f in [Theorem 1](#) with a random function constructed from random vectors. This leads to [Theorem 2](#), our second extension. Although [Theorem 3](#) is a special case of this result, we focus on [Theorem 2](#) in this section to establish the theoretical groundwork, deferring [Theorem 3](#) and the specific statistical application to the next section.

Theorem 2 (Taylor's Theorems for Random Functions and Random Vectors). Let (Ω, Σ_Ω) , $(\mathcal{X}, \mathcal{B}_\mathcal{X})$, $(\mathcal{Y}, \mathcal{B}_\mathcal{Y})$, $(\mathcal{Z}, \mathcal{B}_\mathcal{Z})$, $(\mathcal{U}, \mathcal{B}_\mathcal{U})$, and $(\mathcal{F}, \mathcal{B}_\mathcal{F})$ be measurable spaces, where $\mathcal{U} \subseteq \mathcal{E} = \mathbb{R}^p$, $\mathcal{F} = \mathbb{R}$, $\mathcal{X} \subseteq \mathcal{U}$, $\mathcal{Y} \subseteq \mathcal{E}$, $\mathcal{Z} \subseteq \mathbb{R}^d$, and $m \geq 1$ is an integer. Let $\mathbf{Z} : (\Omega, \Sigma_\Omega) \rightarrow (\mathcal{Z}, \mathcal{B}_\mathcal{Z})$ be a d -dimensional random vector and $f : \mathcal{Z} \times \mathcal{U} \rightarrow \mathcal{F}$ be a bivariate function. Thus, $f(\mathbf{Z}, \mathbf{u})$ is a random function. Suppose the following four conditions hold:

(R-1) $\mathcal{U}$ is an open set.

(R-2) For each $\mathbf{u} \in \mathcal{U}$, $f(\cdot, \mathbf{u})$ is continuous in argument $\mathbf{z}$.

(R-3) For each $\mathbf{z} \in \mathcal{Z}$, $f(\mathbf{z}, \cdot)$ is m th continuously differentiable in argument $\mathbf{u}$ on $\mathcal{U}$.

(R-4) Let $\mathbf{X} : (\Omega, \Sigma_\Omega) \rightarrow (\mathcal{X}, \mathcal{B}_\mathcal{X})$ and $\mathbf{Y} : (\Omega, \Sigma_\Omega) \rightarrow (\mathcal{Y}, \mathcal{B}_\mathcal{Y})$ be p -dimensional random vectors such that $\{\mathbf{X}(\omega) + t\mathbf{Y}(\omega) : t \in [0, 1]\} \subseteq \mathcal{U}$ for all $\omega \in \Omega$.

Then, there exists a p -dimensional random vector $\boldsymbol{\xi} : (\Omega, \Sigma_\Omega) \rightarrow (\mathcal{U}, \mathcal{B}_\mathcal{U})$ such that

$$f(\mathbf{Z}, \mathbf{X} + \mathbf{Y}) = \sum_{k=0}^{m-1} \frac{1}{k!} \left(\sum_{\ell=1}^p Y_\ell \frac{\partial}{\partial u_\ell} \right)^k f(\mathbf{Z}, \mathbf{X}) + \frac{1}{m!} \left(\sum_{\ell=1}^p Y_\ell \frac{\partial}{\partial u_\ell} \right)^m f(\mathbf{Z}, \boldsymbol{\xi}),$$

and $\boldsymbol{\xi}(\omega) \in \{\mathbf{X}(\omega) + t\mathbf{Y}(\omega) : t \in [0, 1]\} \subseteq \mathcal{U}$ for each $\omega \in \Omega$, where u_ℓ and Y_ℓ are the ℓ th components of $\mathbf{u}$ and $\mathbf{Y}$, and the operator is short for

$$\left(\sum_{\ell=1}^p Y_\ell \frac{\partial}{\partial u_\ell} \right)^k = \sum_{i_1, \dots, i_p \in \mathbb{N} : \sum_{\ell=1}^p i_\ell = k} \frac{k!}{i_1! \cdots i_p!} Y_1^{i_1} \cdots Y_p^{i_p} \frac{\partial^k}{\partial u_1^{i_1} \cdots \partial u_p^{i_p}}, \quad k = 0, 1, \dots, m-1.$$

In particular, for $m = 1$, the mean value theorem holds: there exists a p -dimensional random vector $\boldsymbol{\xi} : (\Omega, \Sigma_\Omega) \rightarrow (\mathcal{U}, \mathcal{B}_\mathcal{U})$ such that

$$f(\mathbf{Z}, \mathbf{X} + \mathbf{Y}) - f(\mathbf{Z}, \mathbf{X}) = \nabla_{\mathbf{u}} f(\mathbf{Z}, \boldsymbol{\xi}) \mathbf{Y},$$

and $\boldsymbol{\xi}(\omega) \in \{\mathbf{X}(\omega) + t\mathbf{Y}(\omega) : t \in [0, 1]\} \subseteq \mathcal{U}$ for each $\omega \in \Omega$, where, with respect to the second argument, $\nabla_{\mathbf{u}} f(\mathbf{z}, \mathbf{u}) \triangleq (\partial f(\mathbf{z}, \mathbf{u}) / \partial u_1, \dots, \partial f(\mathbf{z}, \mathbf{u}) / \partial u_p)$ is the gradient (row) vector.

A detailed proof is provided in the [Supplementary Material](#). A key step in our proof involves justifying the Taylor expansion for a generic bivariate function in a stochastic manner. This bivariate function acts as a smooth version of a Carathéodory function: it is measurable in its first argument and smooth (e.g., continuously differentiable) in its second argument. By employing function composition, [Theorem 2](#) generalizes the deterministic function f in [Theorem 1](#) to a random one, where randomness is introduced by evaluating f at a random vector $\mathbf{Z}$ in the first argument. Our theorem may be extended to settings where randomness arises from other sources. Conditions (R-2) and (R-3) replace condition (D-2) to ensure the function f is sufficiently smooth in both arguments. Compared with the classical Taylor's theorem, [Theorem 2](#) successfully establishes a Taylor expansion under the “random in, random out” framework, providing the theoretical basis to guarantee the measurability of intermediate points. A more “off-the-shelf” version, formulated specifically for likelihood functions and estimators, is presented later in [Theorem 3](#).

3. Statistical Applications

The extended Taylor's theorems have broad applicability across various statistical domains, particularly where Taylor-based methods are essential for understanding statistical results or proofs. In this section, we demonstrate how our results enhance comprehension of these topics and address potential technical gaps.

3.1. Deterministic Functions and Random Vectors

When statistical derivations involve deterministic functions and random variables, using [Theorem 1](#) enhances both rigor and clarity. An example illustrating this improvement is the derivation of the Delta method.

Example (the Delta method) (Jiang 2010, Example 4.4). Suppose $g(\cdot)$ is a continuously differentiable multivariate function defined on an open set. Consider a sequence of random vectors $\{\boldsymbol{\xi}_n\}_{n=1}^\infty$ and a random vector $\boldsymbol{\eta}$ such that $a_n(\boldsymbol{\xi}_n - \mathbf{c}) \rightarrow \boldsymbol{\eta}$ in distribution, where $\mathbf{c}$ is a constant vector and $\{a_n\}_{n=1}^\infty$ is a positive sequence diverging to infinity. To establish the Delta method, the classical mean value theorem is applied, yielding

$$g(\boldsymbol{\xi}_n) = g(\mathbf{c}) + \nabla g(\boldsymbol{\zeta}_n)(\boldsymbol{\xi}_n - \mathbf{c}),$$

where “ $\boldsymbol{\zeta}_n$ lies between $\mathbf{c}$ and $\boldsymbol{\xi}_n$ ”. However, since $\boldsymbol{\xi}_n$ is random, the classical mean value theorem does not formally guarantee the measurability of $\boldsymbol{\zeta}_n$, rendering the argument incomplete in probabilistic terms.

All conditions of [Theorem 1](#) are satisfied in this setting. Therefore, we apply it to the deterministic function g , the random vector $\boldsymbol{\xi}_n$, and the deterministic vector $\mathbf{c}$. This establishes that $\boldsymbol{\zeta}_n$ is a random vector lying between $\boldsymbol{\xi}_n$ and $\mathbf{c}$. As a result, probabilistic statements such as $\nabla g(\boldsymbol{\zeta}_n) - \nabla g(\mathbf{c}) = o_p(1)$ become valid, since $\boldsymbol{\zeta}_n$ is random here. This satisfies the conditions required for applying Slutsky's theorem in subsequent asymptotic analysis.

3.2. Random Functions and Random Vectors

One of the most important applications of our main results arises in maximum likelihood estimation, which frequently involves Taylor expansions of random score functions. In this context, both functions and variables are stochastic. Although [Theorem 2](#) suffices to handle this setting, for more readily applicable use in statistical practice, we also present a statistical version, formulated as a special case of [Theorem 2](#), in the following theorem. Before presenting the theorem and these applications, we briefly review likelihood-based parameter estimation, following the notation of Ibragimov and Has'minskii (1981).

Let Ω be the set of elementary outcomes, let $\mathcal{Z} \subseteq \mathbb{R}^d$ represent the set of observation values, and let $\Theta \subseteq \mathbb{R}^p$ represent the parameter set. Assume that (Ω, Σ_Ω) , $(\mathcal{Z}, \mathcal{B}_\mathcal{Z})$, and $(\Theta, \mathcal{B}_\Theta)$ are measurable spaces.

A single observation. Define a d -dimensional random vector $\mathbf{Z} : (\Omega, \Sigma_\Omega) \rightarrow (\mathcal{Z}, \mathcal{B}_\mathcal{Z})$. Suppose its distribution belongs to the parametric family $\{P_\theta, \theta \in \Theta\}$, possessing a dominating σ -finite measure μ (e.g., the counting measure for a discrete random vector or the Lebesgue measure for a continuous random vector). By the Radon-Nikodym theorem, for each $\theta \in \Theta$, there exists

a nonnegative density function $p(\mathbf{z}; \theta) \triangleq dP_\theta(\mathbf{z})/d\mu(\mathbf{z})$ with respect to μ . We emphasize that, by definition, the univariate mapping $p(\mathbf{z}; \theta) : \mathcal{Z} \rightarrow \mathbb{R}$ is the *density function*; and switching the arguments, the univariate mapping $p(\theta; \mathbf{z}) \triangleq p(\mathbf{z}; \theta) : \Theta \rightarrow \mathbb{R}$ is the *likelihood function* (for a single observation). We define a bivariate mapping

$$\ell(\mathbf{z}, \theta) \triangleq \log p(\mathbf{z}; \theta) : \mathcal{Z} \times \Theta \rightarrow \mathbb{R},$$

where, for each $\mathbf{z} \in \mathcal{Z}$, $\ell_{\mathbf{z}}(\theta) \triangleq \ell(\mathbf{z}, \theta)$ serves as the *log-likelihood function* (for a single observation) in argument θ . Replacing deterministic vector $\mathbf{z}$ with random vector $\mathbf{Z}$, we obtain a *random function* $\ell(\mathbf{Z}, \theta) = \log p(\mathbf{Z}; \theta)$.

Statistics and estimators. Suppose $\mathcal{T}_1$ and $\mathcal{T}_2$ are subsets of Euclidean spaces, $(\mathcal{T}_1, \mathcal{B}_{\mathcal{T}_1})$ and $(\mathcal{T}_2, \mathcal{B}_{\mathcal{T}_2})$ are measurable spaces. We define measurable functions $T_1 : (\mathcal{Z}, \mathcal{B}_{\mathcal{Z}}) \rightarrow (\mathcal{T}_1, \mathcal{B}_{\mathcal{T}_1})$ and $T_2 : (\mathcal{Z}, \mathcal{B}_{\mathcal{Z}}) \rightarrow (\mathcal{T}_2, \mathcal{B}_{\mathcal{T}_2})$ as statistics. Thus, the composite mappings $\tilde{T}_1 \triangleq T_1(\mathbf{Z})$ and $\tilde{T}_2 \triangleq T_2(\mathbf{Z})$ are estimators from (Ω, Σ_Ω) to $(\mathcal{T}_1, \mathcal{B}_{\mathcal{T}_1})$ and $(\mathcal{T}_2, \mathcal{B}_{\mathcal{T}_2})$, respectively.

n independent copies. Suppose $\mathbf{Z}_1, \dots, \mathbf{Z}_n$ are n independent copies of d -dimensional random vector $\mathbf{Z} : (\Omega, \Sigma_\Omega) \rightarrow (\mathcal{Z}, \mathcal{B}_{\mathcal{Z}})$. We consider the joint log-likelihood function of $\mathbf{Z}_1, \dots, \mathbf{Z}_n$ as below. We define $\mathcal{Z}^n, \mathcal{B}_{\mathcal{Z}}^n$, and P_θ^n as the n -products $\mathcal{Z} \times \dots \times \mathcal{Z}$, $\mathcal{B}_{\mathcal{Z}} \times \dots \times \mathcal{B}_{\mathcal{Z}}$, and $P_\theta \times \dots \times P_\theta$, respectively. All the measures P_θ^n possess the densities

$$p_n(\mathbf{z}_{1:n}; \theta) \triangleq \frac{dP_\theta^n(\mathbf{z}_{1:n})}{d\mu^n(\mathbf{z}_{1:n})} = p(\mathbf{z}_1; \theta) \times \dots \times p(\mathbf{z}_n; \theta),$$

$$\mathbf{z}_{1:n} \triangleq (\mathbf{z}_1^\top, \dots, \mathbf{z}_n^\top)^\top \in \mathbb{R}^{nd},$$

with respect to the n -product measure $\mu^n \triangleq \mu \times \dots \times \mu$. Thus, we define a bivariate mapping

$$\ell_n(\mathbf{z}_{1:n}, \theta) \triangleq \log p_n(\mathbf{z}_{1:n}; \theta) : \mathcal{Z}^n \times \Theta \rightarrow \mathbb{R},$$

where, for each $\mathbf{z}_{1:n} \in \mathcal{Z}^n$, $\ell_{n, \mathbf{z}_{1:n}}(\theta) \triangleq \ell_n(\mathbf{z}_{1:n}, \theta)$ is the *joint log-likelihood function* in argument θ . By stacking n observations into one, we can define a (nd) -dimensional random vector $\mathbf{Z}_{1:n} \triangleq (\mathbf{Z}_1^\top, \dots, \mathbf{Z}_n^\top)^\top : (\Omega, \Sigma_\Omega) \rightarrow (\mathcal{Z}^n, \mathcal{B}_{\mathcal{Z}}^n)$ satisfying that its log-likelihood function (for the single (nd) -dimensional observation $\mathbf{Z}_{1:n}$) is equivalent to the joint log-likelihood function (for iid d -dimensional observations $\mathbf{Z}_1, \dots, \mathbf{Z}_n$). Replacing deterministic vector $\mathbf{z}_{1:n}$ with random vector $\mathbf{Z}_{1:n}$, we obtain a *random function* $\ell_n(\mathbf{Z}_{1:n}, \theta) = \log p_n(\mathbf{Z}_{1:n}; \theta)$.

With the definitions of n -copied samples and estimators, we now adapt [Theorem 2](#) to a specialized version suited for random joint log-likelihood functions. This version is “ready-to-use” in statistical applications.

Theorem 3 (Taylor’s Theorems for Random Joint Log-Likelihood Functions and Estimators). Let (Ω, Σ_Ω) , $(\mathcal{T}_1, \mathcal{B}_{\mathcal{T}_1})$, $(\mathcal{T}_2, \mathcal{B}_{\mathcal{T}_2})$, $(\mathcal{Z}, \mathcal{B}_{\mathcal{Z}})$, $(\Theta, \mathcal{B}_\Theta)$, and $(\mathcal{F}, \mathcal{B}_{\mathcal{F}})$ be measurable spaces, where $\Theta \subseteq \mathcal{E} = \mathbb{R}^p$, $\mathcal{F} = \mathbb{R}$, $\mathcal{T}_1 \subseteq \Theta$, $\mathcal{T}_2 \subseteq \mathcal{E}$, $\mathcal{Z} \subseteq \mathbb{R}^d$, and $m \geq 1$ is an integer. Let $\mathbf{Z} : (\Omega, \Sigma_\Omega) \rightarrow (\mathcal{Z}, \mathcal{B}_{\mathcal{Z}})$ be a d -dimensional random vector with a nonnegative density function $p(\mathbf{z}; \theta)$. Suppose $\mathbf{Z}_1, \dots, \mathbf{Z}_n$ are n independent copies of $\mathbf{Z}$. Thus, we let $\mathbf{Z}_{1:n} \triangleq (\mathbf{Z}_1^\top, \dots, \mathbf{Z}_n^\top)^\top : (\Omega, \Sigma_\Omega) \rightarrow (\mathcal{Z}^n, \mathcal{B}_{\mathcal{Z}}^n)$ be a (nd) -dimensional random vector with a nonnegative density function $p_n(\mathbf{z}_{1:n}; \theta) = \prod_{i=1}^n p(\mathbf{z}_i; \theta)$, where $\mathbf{z}_{1:n}$

is the realization of $\mathbf{Z}_{1:n}$. Let $\ell_n(\mathbf{z}_{1:n}, \theta) \triangleq \log p_n(\mathbf{z}_{1:n}; \theta) = \sum_{i=1}^n \log p(\mathbf{z}_i; \theta)$ denote a bivariate function. Thus, for all $\mathbf{z}_{1:n} \in \mathcal{Z}^n$, $\ell_{n, \mathbf{z}_{1:n}}(\theta) \triangleq \ell_n(\mathbf{z}_{1:n}, \theta)$ is the joint log-likelihood function of $\mathbf{Z}_1, \dots, \mathbf{Z}_n$. Replacing $\mathbf{z}_{1:n}$ with $\mathbf{Z}_{1:n}$, $\ell_{n, \mathbf{Z}_{1:n}}(\theta) \triangleq \ell_n(\mathbf{Z}_{1:n}, \theta) \triangleq \sum_{i=1}^n \log p(\mathbf{Z}_i; \theta)$ is a random function. Suppose the following four conditions hold:

(L-1) Θ is an open set.

(L-2) For each $\theta \in \Theta$, $\ell_n(\cdot, \theta) \triangleq \log p_n(\cdot; \theta)$ is continuous in argument $\mathbf{z}_{1:n}$.

(L-3) For each $\mathbf{z}_{1:n} \in \mathcal{Z}^n$, $\ell_n(\mathbf{z}_{1:n}, \cdot) \triangleq \log p_n(\mathbf{z}_{1:n}; \cdot)$ is m th continuously differentiable in argument θ on Θ .

(L-4) Let $T_1 : (\mathcal{Z}^n, \mathcal{B}_{\mathcal{Z}}^n) \rightarrow (\mathcal{T}_1, \mathcal{B}_{\mathcal{T}_1})$ and $T_2 : (\mathcal{Z}^n, \mathcal{B}_{\mathcal{Z}}^n) \rightarrow (\mathcal{T}_2, \mathcal{B}_{\mathcal{T}_2})$ be p -dimensional statistics such that $\{\tilde{T}_1(\omega) + t\tilde{T}_2(\omega) : t \in [0, 1]\} \subseteq \Theta$ for all $\omega \in \Omega$, where the estimators $\tilde{T}_1 \triangleq T_1(\mathbf{Z}_{1:n}(\omega)) : \Omega \rightarrow \Theta$ and $\tilde{T}_2 \triangleq T_2(\mathbf{Z}_{1:n}(\omega)) : \Omega \rightarrow \mathcal{E}$.

Then, there exists a p -dimensional random vector $\xi : (\Omega, \Sigma_\Omega) \rightarrow (\Theta, \mathcal{B}_\Theta)$ such that

$$\ell_n(\mathbf{Z}_{1:n}, \tilde{T}_1 + \tilde{T}_2) = \sum_{k=0}^{m-1} \frac{1}{k!} \left(\sum_{\ell=1}^p \tilde{T}_{2,\ell} \frac{\partial}{\partial \theta_\ell} \right)^k \ell_n(\mathbf{Z}_{1:n}, \tilde{T}_1) + \frac{1}{m!} \left(\sum_{\ell=1}^p \tilde{T}_{2,\ell} \frac{\partial}{\partial \theta_\ell} \right)^m \ell_n(\mathbf{Z}_{1:n}, \xi),$$

and $\xi(\omega) \in \{\tilde{T}_1(\omega) + t\tilde{T}_2(\omega) : t \in [0, 1]\} \subseteq \Theta$ for each $\omega \in \Omega$, where θ_ℓ and $\tilde{T}_{2,\ell}$ are the ℓ th components of θ and $\tilde{T}_2$, and the operator is short for

$$\left(\sum_{\ell=1}^p \tilde{T}_{2,\ell} \frac{\partial}{\partial \theta_\ell} \right)^k = \sum_{i_1, \dots, i_p \in \mathbb{N} : \sum_{\ell=1}^p i_\ell = k} \frac{k!}{i_1! \dots i_p!} \tilde{T}_{2,1}^{i_1} \dots \tilde{T}_{2,p}^{i_p} \frac{\partial^k}{\partial \theta_1^{i_1} \dots \partial \theta_p^{i_p}}, \quad k = 0, 1, \dots, m-1.$$

In particular, for $m = 1$, the mean value theorem holds: there exists a p -dimensional random vector $\xi : (\Omega, \Sigma_\Omega) \rightarrow (\Theta, \mathcal{B}_\Theta)$ such that

$$\ell_n(\mathbf{Z}_{1:n}, \tilde{T}_1 + \tilde{T}_2) - \ell_n(\mathbf{Z}_{1:n}, \tilde{T}_1) = \nabla_\theta \ell_n(\mathbf{Z}_{1:n}, \xi) \tilde{T}_2,$$

and $\xi(\omega) \in \{\tilde{T}_1(\omega) + t\tilde{T}_2(\omega) : t \in [0, 1]\} \subseteq \Theta$ for each $\omega \in \Omega$, where, with respect to the second argument, $\nabla_\theta \ell_n(\mathbf{z}_{1:n}, \theta) \triangleq (\partial \ell_n(\mathbf{z}_{1:n}, \theta) / \partial \theta_1, \dots, \partial \ell_n(\mathbf{z}_{1:n}, \theta) / \partial \theta_p)$ is the gradient (row) vector.

To clarify the relationship between measurable spaces, [Theorem 3](#) begins with the necessary measure-theoretic notations, and so does the proof in the [Supplementary Material](#). Specifically, we treat the estimator as a proper composite map from the sample space to the Euclidean observation space, and finally to the Euclidean parameter space, equipping each with its respective σ - or Borel σ -algebra. Once the measurability of the estimator is settled, we apply [Theorem 2](#) to the single-observation case (i.e., $n = 1$). We then extend this to n observations by defining a “vectorized” random vector on the product measurable space.

[Theorem 3](#) presents a statistical version of Taylor’s theorem that incorporates a random objective function, n -copied random

samples, and estimators. Its conditions are generally mild, with the exception of condition (L-4), which requires estimators. This may need to be explicitly verified in some cases. In the example that follows, we illustrate how [Theorem 3](#) facilitates the analysis of classical maximum likelihood estimation.

Example (maximum likelihood estimation) (Kendall and Stuart 1961, chap. 18.16). In the seminal monograph, the asymptotic normality of the maximum likelihood estimator (MLE) is derived using the classical mean value theorem. Specifically, let $\mathbf{Z}_{1:n}$ and $\ell_n(\theta) \triangleq \ell_n(\mathbf{Z}_{1:n}, \theta)$ denote the random sample of size n and the random version of the joint log-likelihood function for any fixed $\mathbf{Z}_{1:n}$, respectively. The classical mean value theorem is applied to the score function $\ell_n^{(1)}(\theta)$, evaluated at the MLE $\hat{\theta}_n \triangleq \hat{\theta}_n(\mathbf{Z}_{1:n})$, around the true parameter θ_0 ,

$$\ell_n^{(1)}(\hat{\theta}_n) = \ell_n^{(1)}(\theta_0) + (\hat{\theta}_n - \theta_0) \ell_n^{(2)}(\theta_n^*),$$

where “ θ_n^* is some value between $\hat{\theta}_n$ and θ_0 .” This step, however, overlooks the randomness of the score function $\ell_n^{(1)}$ and the MLE $\hat{\theta}_n$, leaving the argument incomplete.

We assume that (a) the parameter space Θ is open, (b) $\ell_n^{(1)}(\mathbf{z}_{1:n}, \theta)$ is continuous in argument $\mathbf{z}_{1:n}$ and (c) continuously differentiable in argument θ , and (d) $T_1 = \theta_0$ and $T_2 = \hat{\theta}_n - \theta_0$ are measurable satisfying that $\{(1-t)\theta_0 + t\hat{\theta}_n(\mathbf{Z}_{1:n}(\omega)) : t \in [0, 1]\} \subseteq \Theta$ for all $\omega \in \Omega$. Thus, all assumptions of [Theorem 3](#) are satisfied. We apply it to the random version of the score function $\ell_n^{(1)}$, the MLE $\hat{\theta}_n(\mathbf{Z}_{1:n})$, and the deterministic parameter θ_0 , yielding that there exists a *random vector* θ_n^* such that

$$\begin{aligned} \ell_n^{(1)}(\mathbf{Z}_{1:n}, \hat{\theta}_n(\mathbf{Z}_{1:n})) - \ell_n^{(1)}(\mathbf{Z}_{1:n}, \theta_0) \\ = \ell_n^{(2)}(\mathbf{Z}_{1:n}, \theta_n^*) (\hat{\theta}_n(\mathbf{Z}_{1:n}) - \theta_0), \end{aligned}$$

and $\theta_n^*(\omega) \in \{(1-t)\theta_0 + t\hat{\theta}_n(\mathbf{Z}_{1:n}(\omega)) : t \in [0, 1]\} \subseteq \Theta$ for each $\omega \in \Omega$. We note that the measurability of θ_n^* is crucial in Kendall and Stuart’s (1961) derivation of asymptotic normality of $\hat{\theta}_n$, because, without it, the subsequent analyses, for example, $\Pr \left\{ \lim_{n \rightarrow \infty} \ell_n^{(2)}(\theta_n^*) = -E \left[\ell_n^{(1,2)}(\theta_0) \right] \right\} = 1$, may not be reasonable within a probabilistic framework.

In the next two examples, we demonstrate how [Theorem 3](#) can aid in analyzing random functions beyond the classical likelihoods.

Example (M-estimation) (van der Vaart 1998, chap. 5.3). Let $\mathbf{Z}_{1:n}$, $\hat{\theta}_n \triangleq \hat{\theta}_n(\mathbf{Z}_{1:n})$, and $\Psi_n(\theta) \triangleq \Psi_n(\mathbf{Z}_{1:n}, \theta) = 0$ denote the random sample, the M -estimator, and a system of random equations (e.g., the average of score equations in maximum likelihood estimation) for any fixed $\mathbf{Z}_{1:n}$. In providing the asymptotic normality of M -estimators, van der Vaart (1998) expands $\Psi_n(\hat{\theta}_n)$ using a classical version of the Taylor series around the true value θ_0 and obtains that

$$\begin{aligned} 0 = \Psi_n(\hat{\theta}_n) = \Psi_n(\theta_0) + (\hat{\theta}_n - \theta_0) \Psi_n^{(1)}(\theta_0) \\ + \frac{1}{2} (\hat{\theta}_n - \theta_0)^2 \Psi_n^{(2)}(\tilde{\theta}_n), \end{aligned}$$

where “ $\tilde{\theta}_n$ is a point between $\hat{\theta}_n$ and θ_0 .” Obviously, the classical Taylor’s theorem fails due to a lack of consideration of the randomness of Ψ_n and $\hat{\theta}_n$.

Under assumptions that (a) the parameter space is an open set, (b) $\Psi_n(\mathbf{z}_{1:n}, \theta)$ is continuous in argument $\mathbf{z}_{1:n}$ and

(c) twice continuously differentiable in argument θ , and (d) $T_1 = \theta_0$ and $T_2 = \hat{\theta}_n - \theta_0$ are measurable satisfying that $\{(1-t)\theta_0 + t\hat{\theta}_n(\mathbf{Z}_{1:n}(\omega)) : t \in [0, 1]\} \subseteq \Theta$ for all $\omega \in \Omega$. All assumptions of [Theorem 3](#) are satisfied. Applying [Theorem 3](#), we still obtain the above expansion while $\tilde{\theta}_n$ is a *random variable* between $\hat{\theta}_n$ and θ_0 . Thus, it is reasonable to verify that $\Psi_n^{(2)}(\tilde{\theta}_n) = O_P(1)$ in subsequent analysis for the asymptotic normality of M -estimators.

Example (profile estimation) (Kosorok 2008, chap. 19.3). Let $\{P_{\theta, \eta} : \theta \in \Theta, \eta \in \mathcal{H}\}$ denote a family of semiparametric models, where $\Theta \subseteq \mathbb{R}^p$ is a finite-dimensional set and $\mathcal{H}$ is an infinite-dimensional set. Let $\mathbf{Z}_{1:n}$ and $\ell_n(\theta, \eta) \triangleq \ell_n(\mathbf{Z}_{1:n}, \theta, \eta)$ denote the random sample of size n and the random version of the joint log-likelihood function for any fixed $\mathbf{Z}_{1:n}$. Additionally, define θ_0 and η_0 as the true parameters.

Following the Kosorok’s (2008) notation, let $\hat{\eta}_{\theta_0} \triangleq \arg\max_{\eta \in \mathcal{H}} \ell_n(\theta_0, \eta)$, let $\{\tilde{\theta}_n\}_{n=1}^\infty$ denote a sequence of estimators

converging to the true value θ_0 in probability, and let $\tilde{\psi} \triangleq (\theta_0, \hat{\eta}_{\theta_0})$. We assume there exists a map $t \mapsto \eta_t(\tilde{\psi}) \triangleq \eta_{\tilde{\psi}, t}$ from the neighborhood of θ_0 into the parameter set for $\hat{\eta}_{\theta_0}$, satisfying that $\eta_{\tilde{\psi}, t} \in \hat{\mathcal{H}}$ if $\|t - \theta_0\|$ is sufficiently small, and $\eta_{\tilde{\psi}, t} = \hat{\eta}_{\theta_0}$ if $\tilde{\psi} = (\theta_0, \hat{\eta}_{\theta_0}) \in \Theta \times \hat{\mathcal{H}}$, where $\hat{\mathcal{H}}$ is a suitable enlargement of $\mathcal{H}$. Thus, we rewrite $\eta_{\tilde{\psi}, \theta_0} \triangleq \hat{\eta}_{\theta_0}$ using $t = \theta_0$ and define $\eta_{\tilde{\psi}, \tilde{\theta}_n}$ using $t = \tilde{\theta}_n$ (which is sufficiently close to θ_0). Thus, Kosorok (2008) applies the classical Taylor theorem and obtains that

$$\begin{aligned} \ell_n(\tilde{\theta}_n, \eta_{\tilde{\psi}, \tilde{\theta}_n}) = \ell_n(\theta_0, \eta_{\tilde{\psi}, \theta_0}) + (\tilde{\theta}_n - \theta_0)^\top \ell_n^{(1)}(\theta_0, \eta_{\tilde{\psi}, \theta_0}) \\ + \frac{1}{2} (\tilde{\theta}_n - \theta_0)^\top \ell_n^{(2)}(\tilde{t}_n, \eta_{\tilde{\psi}, \tilde{t}_n}) (\tilde{\theta}_n - \theta_0), \end{aligned}$$

for “some $\tilde{t}_n$ on the line segment between $\tilde{\theta}_n$ and θ_0 .” Unfortunately, the conclusion may be incomplete because the randomness of the function ℓ_n and the estimators $\tilde{\theta}_n$, $\eta_{\tilde{\psi}, \tilde{\theta}_n}$, and $\eta_{\tilde{\psi}, \theta_0}$ violates the assumptions in the classical Taylor theorem.

Given $\tilde{\psi} \triangleq \tilde{\psi}(\mathbf{Z}_{1:n}) \triangleq (\theta_0, \hat{\eta}_{\theta_0}(\mathbf{Z}_{1:n}))$ and the map $t \mapsto \eta_{\tilde{\psi}, t} \triangleq \eta_{\tilde{\psi}(\mathbf{Z}_{1:n}), t}$, we rewrite

$$\ell_n(\mathbf{Z}_{1:n}, \theta, \eta_{\tilde{\psi}(\mathbf{Z}_{1:n}), \theta}) \triangleq \ell_{\tilde{\psi}, n}(\mathbf{Z}_{1:n}, \theta)$$

for θ sufficiently close to θ_0 . Suppose the following assumptions hold: (a) the parameter space Θ is open, (b) $\ell_{\tilde{\psi}, n}(\mathbf{z}_{1:n}, \theta)$ is continuous in argument $\mathbf{z}_{1:n}$ and (c) is twice continuously differentiable in argument θ , and (d) $T_1 = \theta_0$ and $T_2 = \tilde{\theta}_n - \theta_0$ are measurable satisfying that $\{(1-t)\theta_0 + t\tilde{\theta}_n(\mathbf{Z}_{1:n}(\omega)) : t \in [0, 1]\} \subseteq \Theta$ for all $\omega \in \Omega$. Applying [Theorem 3](#), we obtain that

$$\begin{aligned} \ell_{\tilde{\psi}, n}(\tilde{\theta}_n) = \ell_{\tilde{\psi}, n}(\theta_0) + (\tilde{\theta}_n - \theta_0)^\top \ell_{\tilde{\psi}, n}^{(1)}(\theta_0) \\ + \frac{1}{2} (\tilde{\theta}_n - \theta_0)^\top \ell_{\tilde{\psi}, n}^{(2)}(\tilde{t}_n) (\tilde{\theta}_n - \theta_0), \end{aligned}$$

where $\tilde{t}_n$ is a *random vector* between $\tilde{\theta}_n$ and θ_0 . This implies that $n^{-1} \ell_{\tilde{\psi}, n}^{(2)}(\tilde{t}_n) = n^{-1} \ell_n^{(2)}(\tilde{t}_n, \eta_{\tilde{\psi}, \tilde{t}_n}) \rightarrow P_0 \ell^{(2)}(\theta_0, \eta_0)$ in probability, where $\ell(\theta, \eta)$ denotes the random version of the log-likelihood function for a single observation, and P_0 is the probability measure at the true value.

Analogous modifications help resolve arguments concerning the limiting behavior of MLEs in both the single-parameter case (Cox and Hinkley 1974; Bhattacharya, Lin, and Patrangenaru 2016) and the general multi-parameter case (Kendall and Stuart 1961; Jiang 2010). Additionally, by applying Theorem 3, we can address measurability gaps in the estimation of normal probabilities (Lehmann 2004, chap. 4.2), in the derivation of the Cramér-Rao information inequality (Bhattacharya, Lin, and Patrangenaru 2016, chap. 4.3), and in the estimation procedure of the jackknife variance (Shao 2003, chap. 5.5). More broadly, our results extend to various areas, including the asymptotic properties of likelihood ratio statistics (van der Vaart 1998, chap. 16.2), the analysis of convergence rates for penalized M -estimators (Kosorok 2008, chap. 21.5).

4. Discussion

We have developed statistical extensions of Taylor's theorems and the mean value theorems, specifically for settings involving both random functions and random variables. We rigorously establish the measurable version of Taylor's theorems, explicitly accounting for the randomness inherent in the functions. Extending beyond classical results, our theorems guarantee the existence of a random intermediate point between two given random vectors. Additionally, we offer a sufficient condition for measurability: in practice, continuity of the highest-order derivatives is typically sufficient, while theoretically, this can be relaxed to quasi-continuity and closed graph conditions. We adapt the main results to a statistical setting, providing a formulation that is readily applicable to statistical problems. Our work represents a refined technical advancement that enhances clarity and simplifies statistical reasoning, particularly in the study of the asymptotic behavior of parametric, semiparametric, and nonparametric estimators. These results contribute to making statistical arguments more concise and accessible.

Supplementary Materials

The Supplementary Material contains the classical Taylor's theorems, additional definitions, technical lemmas, and proofs of our main results.

Acknowledgments

We appreciate the time and effort that the editor, the associate editor, and the reviewers have dedicated to providing their valuable feedback on our article.

Disclosure Statement

No potential competing interest was reported by the authors.

Funding

This work was partially supported by the National Heart, Lung, and Blood Institute of the National Institutes of Health under Grant Number 1R01HL175410.

ORCID

Yifan Yang  <https://orcid.org/0000-0001-5727-6540>

Ming Wang  <https://orcid.org/0000-0002-1959-4879>

References

- Aliprantis, C. D., and Border, K. C. (2006), *Infinite Dimensional Analysis* (3rd ed.), Berlin: Springer. A hitchhiker's guide. [2]
- Benichou, J., and Gail, M. H. (1989), "A Delta Method for Implicitly Defined Random Variables," *The American Statistician*, 43, 41–44. [2]
- Bhattacharya, R., Lin, L., and Patrangenaru, V. (2016), *A Course in Mathematical Statistics and Large Sample Theory*, Springer Texts in Statistics, New York: Springer-Verlag. [7]
- Cox, D. R., and Hinkley, D. V. (1974), *Theoretical Statistics*, London: Chapman and Hall. [7]
- Di Crescenzo, A. (1999), "A Probabilistic Analogue of the Mean Value Theorem and its Applications to Reliability Theory," *Journal of Applied Probability*, 36, 706–719. [2]
- Dindoš, M., and Toma, V. (1997), "Filippov Implicit Function Theorem for Quasi-Carathéodory Functions," *Journal of Mathematical Analysis and Applications*, 214, 475–481. [3]
- Feng, C., Wang, H., Han, Y., Xia, Y., and Tu, X. M. (2013), "The Mean Value Theorem and Taylor's Expansion in Statistics," *The American Statistician*, 67, 245–248. [3]
- Himmelberg, C. J. (1975), "Measurable Relations," *Fundamenta Mathematicae*, 87, 53–72. [3]
- Ibragimov, I. A., and Has'minskii, R. Z. (1981), *Statistical Estimation*, volume 16 of Applications of Mathematics, New York-Berlin: Springer-Verlag. Asymptotic theory, Translated from the Russian by Samuel Kotz. [4]
- Jiang, J. (2010), *Large Sample Techniques for Statistics*, Springer Texts in Statistics, New York: Springer. [4,7]
- Kendall, M. G., and Stuart, A. (1961), *The Advanced Theory of Statistics. Vol. 2: Inference and Relationship* (3rd ed.), New York: Hafner Publishing Co. [6,7]
- Kosorok, M. R. (2008), *Introduction to Empirical Processes and Semiparametric Inference*, Springer Series in Statistics, New York: Springer. [6,7]
- Lang, S. (1993), *Real and Functional Analysis* (3rd ed.), volume 142 of Graduate Texts in Mathematics, New York: Springer-Verlag. [1,2]
- Lehmann, E. L. (2004), *Elements of Large-Sample Theory*, Springer Texts in Statistics, New York: Springer-Verlag. [7]
- Lin, G. D. (1994), "On a Probabilistic Generalization of Taylor's Theorem," *Statistics & Probability Letters*, 19, 239–243. [2]
- Massey, W. A., and Whitt, W. (1993), "A Probabilistic Generalization of Taylor's Theorem," *Statistics & Probability Letters*, 16, 51–54. [2]
- Rudin, W. (1976), *Principles of Mathematical Analysis* (3rd ed.), International Series in Pure and Applied Mathematics. New York-Auckland-Düsseldorf: McGraw-Hill Book Co. [3]
- Shao, J. (2003), *Mathematical Statistics* (2nd ed.), Springer Texts in Statistics. New York: Springer-Verlag. [7]
- Tao, T. (2011), *An Introduction to Measure Theory*, volume 126 of Graduate Studies in Mathematics, Providence, RI: American Mathematical Society. [3]
- Tao, T. (2022), *Analysis II* (4th ed.), volume 38 of Texts and Readings in Mathematics, Singapore: Springer; New Delhi: Hindustan Book Agency. [3]
- Taylor, B. (1717), *Methodus incrementorum directa & inversa* (2nd ed.). London. [1]
- van der Vaart, A. W. (1998), *Asymptotic Statistics*, volume 3 of Cambridge Series in Statistical and Probabilistic Mathematics, Cambridge: Cambridge University Press. [6,7]